

YEAR 8

- 1) In a right triangle, the length of the hypotenuse is 18m and the length of one of the side is 15m. Find the length of the other side. Round your answer to the nearest tenth.
- 2) Write an equation for this situation.

Patricia has **P** posters. She sold 8 and has 18 left.

The surface area of a cube is 216cm^2 (Question 3-4)

- 3) What is the surface area of one face of the cube?
- 4) What is the length of one edge of the cube?
- 5) The original price for a pair of jeans was £48.00. On the first day, the price was decreased 15%. On the second day, the price was reduced an additional 20% off the original price. What was the sale price on the second day?
- 6) The ordered pair (5,) is in the linear relation with equation $Y = -2x + 8$. Find the missing number in ordered pair.

Solve these equations. (Question 7-8)

- 7) $9 + \frac{d}{4} = 23$
- 8) $\frac{x}{-6} = -9$
- 9) Calculate the sale price of this item before taxes. 30% off a bike for £397.95
- 10) The table below shows the values for the linear relation with equation $y = b - x$, where b is a constant. Find the value of b

X	1	2	3	4	5
y	11	10	9	8	7

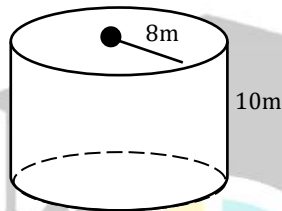
- 11) In 3 stores, the same detergent is on special. Which store offers the best buy?
Explain.

Store	A:	6 bottles for	£12.48
Store	B:	7 bottles for	£14.42
Store	C:	5 bottles for	£10.35

On the stock market, the price of one share of Elizalde electronics dropped by a total of £1.25 over a period of five business days. The price of one share of Stratos Metals steel dropped by a total of £15 during the same five days. Meanwhile, the price of one share of World style packaging increased by a total of £8 over the five days. (Question 12-15)

Solve:

- 12) What was the average amount of money lost in one day of 30 shares of Elizalde Electronics?
- 13) What was the average amount of money lost in one day for 25 shares of Stratos Metals Steel?
- 14) What was the average amount of money gained in one day for 10 shares of World Style Packaging?
- 15) Suppose you were of the 30 shares of Elizalde electronics, 25 shares of Stratos Metals Steel, and 10 shares of World Style Packaging. What was the average amount of money lost or gained in one day for your entire stock portfolio?
- 16) Find the surface area of this cylinder to the nearest square meter.



- 17) A waste basket shaped like a triangular prism has a volume of 10368cm^3 . its triangular base is 24cm wide and 18cm high. What is the height of the waste basket?
- 18) Deon buys a laptop for £8000. The next year he sells it for £6900. Does he make a profit or a loss? Show your workings out.

It cost £5.00 to process a roll of film. David has a number of rolls of film and wants to know the total cost of developing them. (Question 19-20)

- 19) Write an equation in the form of $ax = b$ to represent the problem.
- 20) What is the cost of developing 13 rolls of film?